

SIMATS ENGINEERING
CSA0921 - PROGRAMMING IN JAVA
CO4 - AT1 - Industry Problem Solving

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AIM: To develop a Swing-based Railway Reservation Applet for searching trains, booking tickets, and cancelling reservations using combo boxes, tables, buttons, event listeners, input validation, and seat availability management.

CODE :

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.HashMap;

public class RailwayReservationApplet extends JApplet
    implements ActionListener {

    JComboBox<String> fromBox, toBox;
    JTextField dateField;
    JTextField passengerField, seatsField, pnrField;

    JButton searchButton;
    JButton bookButton, cancelButton, clearButton;

    JTable trainTable;
    DefaultTableModel model;

    JTextArea outputArea;

    HashMap<String, Integer> seatAvailability =
        new HashMap<String, Integer>();
```

```
public void init() {

    seatAvailability.put("101", 120);
    seatAvailability.put("106", 85);
    seatAvailability.put("110", 64);
    seatAvailability.put("129", 42);

    setSize(900, 650);

    JPanel mainPanel =
        new JPanel(new BorderLayout(10, 10));

    mainPanel.setBorder(
        BorderFactory.createEmptyBorder(
            10, 10, 10, 10));

    // TITLE
    JLabel title = new JLabel(
        "RAILWAY RESERVATION SYSTEM",
        JLabel.CENTER);

    title.setFont(
        new Font("Arial", Font.BOLD, 24));

    mainPanel.add(title, BorderLayout.NORTH);

    // SEARCH PANEL
    JPanel searchPanel =
        new JPanel(new GridLayout(2, 4, 10, 10));

    searchPanel.setBorder(
        BorderFactory.createTitledBorder(
            "Search Trains"));

    fromBox = new JComboBox<String>(
        new String[]{
            "Select",
```

```
        "Chennai",  
        "Bangalore",  
        "Coimbatore",  
        "Madurai",  
        "Trichy"  
    });
```

```
toBox = new JComboBox<String>(  
    new String[] {  
        "Select",  
        "Chennai",  
        "Bangalore",  
        "Coimbatore",  
        "Madurai",  
        "Trichy"  
    });
```

```
dateField = new JTextField();
```

```
searchButton =  
    new JButton("Search Trains");
```

```
searchPanel.add(new JLabel("From Station:"));  
searchPanel.add(fromBox);
```

```
searchPanel.add(new JLabel("To Station:"));  
searchPanel.add(toBox);
```

```
searchPanel.add(new JLabel("Travel Date:"));  
searchPanel.add(dateField);
```

```
searchPanel.add(new JLabel(""));  
searchPanel.add(searchButton);
```

```
// TRAIN TABLE
```

```
String[] columns = {  
    "Train No",
```

```
        "Train Name",
        "From",
        "To",
        "Available Seats"
    };

    model =
        new DefaultTableModel(columns, 0);

    trainTable = new JTable(model);

    trainTable.setSelectionMode(
        ListSelectionModel.SINGLE_SELECTION);

    JScrollPane tableScrollPane =
        new JScrollPane(trainTable);

    // BOOKING PANEL
    JPanel bookingPanel =
        new JPanel(new GridLayout(3, 2, 10, 10));

    bookingPanel.setBorder(
        BorderFactory.createTitledBorder(
            "Booking / Cancellation"));

    passengerField = new JTextField();
    seatsField = new JTextField();
    pnrField = new JTextField();

    bookingPanel.add(
        new JLabel("Passenger Name:"));
    bookingPanel.add(passengerField);

    bookingPanel.add(
        new JLabel("Number of Seats:"));
    bookingPanel.add(seatsField);
```

```
bookingPanel.add(
    new JLabel("PNR:"));
bookingPanel.add(pnrField);

// BUTTON PANEL
JPanel buttonPanel =
    new JPanel(new FlowLayout());

bookButton =
    new JButton("Book Ticket");

cancelButton =
    new JButton("Cancel Ticket");

clearButton =
    new JButton("Clear");

bookButton.addActionListener(this);
cancelButton.addActionListener(this);
clearButton.addActionListener(this);
searchButton.addActionListener(this);

buttonPanel.add(bookButton);
buttonPanel.add(cancelButton);
buttonPanel.add(clearButton);

// OUTPUT AREA
outputArea = new JTextArea(7, 50);

outputArea.setEditable(false);

JScrollPane outputScrollPane =
    new JScrollPane(outputArea);

JPanel outputPanel =
    new JPanel(new BorderLayout());
```

```
outputPanel.setBorder(  
    BorderFactory.createTitledBorder(  
        "Booking Details / Output"));  
  
outputPanel.add(  
    outputScrollPane,  
    BorderLayout.CENTER);  
  
// CENTER PANEL  
JPanel centerPanel =  
    new JPanel(new BorderLayout(10, 10));  
  
JPanel topPanel =  
    new JPanel(new BorderLayout());  
  
topPanel.add(  
    searchPanel,  
    BorderLayout.NORTH);  
  
centerPanel.add(  
    topPanel,  
    BorderLayout.NORTH);  
  
centerPanel.add(  
    tableScrollPane,  
    BorderLayout.CENTER);  
  
JPanel bottomPanel =  
    new JPanel(new BorderLayout());  
  
bottomPanel.add(  
    bookingPanel,  
    BorderLayout.CENTER);  
  
bottomPanel.add(  
    buttonPanel,  
    BorderLayout.SOUTH);
```

```

        centerPanel.add(
            bottomPanel,
            BorderLayout.SOUTH);

        mainPanel.add(
            centerPanel,
            BorderLayout.CENTER);

        mainPanel.add(
            outputPanel,
            BorderLayout.SOUTH);

        add(mainPanel);

        displayTrains();
    }

// DISPLAY TRAINS
public void displayTrains() {

    model.setRowCount(0);

    model.addRow(new Object[] {
        "101",
        "Chennai Express",
        "Chennai",
        "Bangalore",
        seatAvailability.get("101")
    });

    model.addRow(new Object[] {
        "106",
        "Kaveri Express",
        "Chennai",
        "Bangalore",
        seatAvailability.get("106")
    });
}

```

```

    });

    model.addRow(new Object[] {
        "110",
        "Brindavan Express",
        "Chennai",
        "Bangalore",
        seatAvailability.get("110")
    });

    model.addRow(new Object[] {
        "129",
        "Bangalore Mail",
        "Chennai",
        "Bangalore",
        seatAvailability.get("129")
    });
}

// BUTTON EVENTS
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == searchButton) {
        searchTrain();
    }

    else if (e.getSource() == bookButton) {
        bookTicket();
    }

    else if (e.getSource() == cancelButton) {
        cancelTicket();
    }

    else if (e.getSource() == clearButton) {
        clearFields();
    }
}

```



```
}
```

```
// SEARCH TRAIN
```

```
public void searchTrain() {
```

```
    String from =  
        fromBox.getSelectedItem().toString();
```

```
    String to =  
        toBox.getSelectedItem().toString();
```

```
    if (from.equals("Select") ||  
        to.equals("Select")) {
```

```
        JOptionPane.showMessageDialog(  
            this,  
            "Please select From and To stations.");
```

```
        return;  
    }
```

```
    if (from.equals(to)) {
```

```
        JOptionPane.showMessageDialog(  
            this,  
            "From and To stations cannot be same.");
```

```
        return;  
    }
```

```
    if (!validateDate()) {  
        return;  
    }
```

```
    model.setRowCount(0);
```

```
    if (from.equals("Chennai") &&
```

```

        to.equals("Bangalore")) {

displayTrains();

outputArea.setText(
    "Search Result Found:\n" +
    "From: " + from + "\n" +
    "To: " + to + "\n" +
    "Travel Date: " +
    dateField.getText());
}

else {

    outputArea.setText(
        "No trains available for the selected route.");
    }
}

// DATE VALIDATION
public boolean validateDate() {

String date =
    dateField.getText().trim();

if (date.isEmpty()) {

    JOptionPane.showMessageDialog(
        this,
        "Please enter travel date.");

    return false;
}

try {

    SimpleDateFormat sdf =

```

```

        new SimpleDateFormat("dd/MM/yyyy");

sdf.setLenient(false);

Date travelDate =
    sdf.parse(date);

if (travelDate.before(new Date())) {

    JOptionPane.showMessageDialog(
        this,
        "Travel date cannot be in the past.");

    return false;
}

}

catch (Exception e) {

    JOptionPane.showMessageDialog(
        this,
        "Enter date in DD/MM/YYYY format.");

    return false;
}

return true;
}

// BOOK TICKET
public void bookTicket() {

    int row =
        trainTable.getSelectedRow();

    if (row < 0) {

```

```

        JOptionPane.showMessageDialog(
            this,
            "Please select a train.");

        return;
    }

    String passenger =
        passengerField.getText().trim();

    if (passenger.isEmpty()) {

        JOptionPane.showMessageDialog(
            this,
            "Please enter passenger name.");

        return;
    }

    int seats;

    try {

        seats = Integer.parseInt(
            seatsField.getText().trim());

        if (seats <= 0)
            throw new Exception();

    }

    catch (Exception e) {

        JOptionPane.showMessageDialog(
            this,
            "Enter a valid number of seats.");
    }

```

```

        return;
    }

    String trainNo =
        model.getValueAt(row, 0).toString();

    int available =
        seatAvailability.get(trainNo);

    if (seats > available) {

        JOptionPane.showMessageDialog(
            this,
            "Only " + available +
            " seats are available.");

        return;
    }

    available -= seats;

    seatAvailability.put(
        trainNo, available);

    String pnr =
        "PNR" +
        (1000 + (int)(Math.random() * 9000));

    pnrField.setText(pnr);

    String trainName =
        model.getValueAt(row, 1).toString();

    String from =
        model.getValueAt(row, 2).toString();

```

```

String to =
    model.getValueAt(row, 3).toString();

displayTrains();

outputArea.setText(
    "Ticket Booked Successfully!\n\n" +
    "PNR Number    : " + pnr + "\n" +
    "Passenger Name : " + passenger + "\n" +
    "Train Name     : " + trainName + "\n" +
    "From - To      : " + from +
    " --> " + to + "\n" +
    "Travel Date    : " +
    dateField.getText() + "\n" +
    "Seats Booked   : " + seats);

JOptionPane.showMessageDialog(
    this,
    "Ticket booked successfully!\n" +
    "PNR: " + pnr);
}

// CANCEL TICKET
public void cancelTicket() {

    String pnr =
        pnrField.getText().trim();

    if (pnr.isEmpty()) {

        JOptionPane.showMessageDialog(
            this,
            "Please enter PNR number.");

        return;
    }
}

```

```
outputArea.setText(
    "Cancellation Request\n\n" +
    "PNR Number : " + pnr + "\n" +
    "Ticket cancellation processed successfully.");

JOptionPane.showMessageDialog(
    this,
    "Ticket cancelled successfully.");
}

// CLEAR
public void clearFields() {

    fromBox.setSelectedIndex(0);
    toBox.setSelectedIndex(0);

    dateField.setText("");
    passengerField.setText("");
    seatsField.setText("");
    pnrField.setText("");

    outputArea.setText("");

    displayTrains();
}
}
```

OUTPUT:

Railway Reservation System

RAILWAY RESERVATION SYSTEM

Search Trains

From:Chennai

To:Bangalore

Travel Date (dd/MM/yyyy):25/08/2026

Search Trains

Available Trains

Train No	Train Name	From	To	Available Seats
101	Chennai Express	Chennai	Bangalore	120
106	Kaveri Express	Chennai	Bangalore	85
110	Brindavan Express	Chennai	Bangalore	64
129	Bangalore Mail	Chennai	Bangalore	42

Booking / Cancellation

Passenger Name:Vidhushini T S

Number of Seats:2

PNR (for cancellation):

Book Ticket

Cancel Ticket

Clear

Booking Details / Output

Ticket Booked Successfully!
PNR: PNR5274
Passenger Name: Vidhushini T S
Train Name: Chennai Express
From - To: Chennai --> Bangalore
Travel Date: 25/08/2026
Seats Booked: 2